Wendy Q. Sun

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EDUCATION

Massachusetts Institute of Technology — Cambridge, MA
Double major: B.S. in Physics — 2022 - 2026
B.S. in Artificial Intelligence & Decision Making
Minor in Mathematics
GPA : 4.9/5.0

HONORS

Phi Beta Kappa Honor Society (top 10% of the graduating class) — 2026
Sigma Pi Sigma Physics and Astronomy Honor Society — 2026
Tau Beta Pi Engineering Honor Society — 2026

PUBLICATIONS

3 primary author papers, ADS library. 16 total papers, ADS library.
† marks equal contribution.

Primary Author Papers

3. **Wendy Q. Sun**, Rohan P. Naidu, Hanpy Liu, Anna de Graaff, Jenny E. Greene, Jorryt Matthee, et al., *Overmassive No More: The Case for Little Red Dots Hosting Black Hole Seeds as Massive as Single Supermassive Stars*, submitted to OJAp.
2. **Wendy Q. Sun**, Rohan P. Naidu, Jorryt Matthee, Anna de Graaff, et al., *Little Red Dot − Host Galaxy = Black Hole Star: A Gas-Enshrouded Heart at the Center of Every Little Red Dot*, OJAp, 9, 62505, 2026 [48 citations].
1. C. Loughridge†, **Q. Sun**†, et al., *DafnyBench: A Benchmark for Formal Software Verification*, The 4th Workshop on Mathematical Reasoning and AI at NeurIPS, 2024 [104 citations].

Contributing Author Papers

13. G. Limberg et al., *Chemodynamical Analysis of a CNO-Enhanced Ultra Metal-poor Star ([Fe/H] < −4): Insights into Early Enrichment by Faint Population III Supernovae*, submitted to AAS journals.
12. C. Ashall et al., *Can Little Red Dots Contribute To The Early Universe Cosmic Dust Budget?*, submitted to ApJ.
11. R. P. Naidu et al., *A gas-enshrouded and gas-reddened black hole at cosmic dawn*, Nature, 656, 329, 2026.
10. Y. Ishikawa et al., *Spatial decomposition of Little Red Dots with JWST/NIRSpec IFU into broad-line red cores and narrow-line blue host galaxies*, arXiv.
9. R. P. Naidu et al., *Little Red Dots as Intermediate Mass, Super-Eddington Engines: Insights from Type IIn Supernovae and The 1837–1856 Great Eruption of η Carinae*, arXiv.
8. A. Weibel, R. P. Naidu, et al., *Black Hole Stars Across the Universe: Identifying Central Engine Dominated Little Red Dots at $z \sim 1.5 - 9.5$*, arXiv.
7. B. Wang et al., *Water absorption confirms cool atmospheres in two little red dots*, arXiv.
6. A. Torralba et al., *A Black Hole Star at Cosmic Noon: Extreme Balmer break, photospheric continuum, and broad absorption by thick winds in a Little Red Dot at $z = 1.7$*, ApJL, 1005, 37, 2026.
5. Z. Liu, R. P. Naidu, A. Secunda, et al., *How I Wonder What You Are – JWST's Little Red Dots do not TWINKLE*, submitted to ApJ.

4. T. Li et al., *Fractal Generative Models*, Transactions of Machine Learning Research (TMLR), 2025.
3. J. C. Carr et al., *Focused Skill Discovery: Learning to Control Specific State Variables while Minimizing Side Effects*, The 2nd Reinforcement Learning Conference (RLC), 2025.
2. S. Casper, C. Ezell, et al., *Black-Box Access is Insufficient for Rigorous AI Audits*, The ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2024.
1. W. Gurnee et al., *Universal Neurons in GPT2 Language Models*, Transactions of Machine Learning Research (TMLR), 2024.

PRESENTATIONS

ISSI Bern, *LRD − Host Galaxy = BH*: A Gas-Enshrouded Heart at the Center of Every LRD*
Meeting, Feb. 2026

RESEARCH EXPERIENCE

Weighing Central Engines of Little Red Dots Honolulu, HI
Institute for Astronomy, University of Hawai'i at Mānoa • *Advisor: Rohan Naidu* July 2026 - present

· Fit stellar atmosphere models to little red dot central engines (black hole stars) and estimated their masses with four non-virial methods, finding masses consistent with black hole stars arising from single supermassive stars.

Host Galaxy Separation for Little Red Dots Cambridge, MA
MIT Kavli Institute • *Advisor: Rohan Naidu* July 2025 - May 2026

· Decomposed 90+ little red dots from the DAWN JWST Archive into black hole star and host galaxy components, finding population-level evidence that every little red dot hosts a black hole star.

Fractal Generative Models Cambridge, MA
MIT CSAIL, He Vision Group • *Advisors: Tianhong Li, Kaiming He* Nov. 2024 - Feb. 2025

· Contributed to building and validating fractal generative models that abstract generative models into atomic generative modules.

DafnyBench: A Benchmark for Formal Software Verification Cambridge, MA
NSF IAIFI, Tegmark AI Safety Group • *Advisor: Max Tegmark* Feb. - Nov. 2024

· Built DafnyBench, the largest formal software verification benchmark at the time (750+ programs), and evaluated large language models' capability to write formally verifiable code.

· Built Dafny Autopilot, a VS Code extension that leverages GPT-4o or Claude 3.5 Sonnet to verify AI-generated code (400+ installs).

Focused Skill Discovery: Using Per-Factor Empowerment to Control State Variables Berkeley, CA
Center for Human-Compatible AI, UC Berkeley • *Advisor: Cam Allen* June - Aug. 2024

· Investigated focused skill discovery in reinforcement learning, where agents learn skills that change one state variable at a time to explore more efficiently and avoid side effects.

RELEVANT COURSEWORK

Physics	Astrophysics I (grad), General Relativity (grad), Statistical Mechanics I (grad), Relativistic Quantum Field Theory I (grad), Electromagnetic Theory (grad)
Mathematics	Real Analysis, Numerical Analysis, Theory of Computation, Combinatorial Optimization